MERSENNES EXPONENTS SEQUENCES LATTICES in TERMS nx2 and nx2+1

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Abstract

Mersennes Exponents can be calculate using a recursive sequences from two fundamental latices with base M1= 2 and M2= 3 which are the first two Mersennes exponents.

Using the pattern 0101...etc where 0 is n**x2** (multiply by 2) and 1 is n**x2+1,** I calculate all existing known Mersennes exponents and here is the study results.

Using an algorithm app (github) I generate 50 recursive sequences latices for the all 52 Mersennes Exponents in form of:

Lattice 10 (Pattern: **11111**)
Contains duplicates
3 → 7 (×2+1) → 15 (×2+1) → 31 (×2+1) → 63 (×2+1) → **127** (×2+1)

Lattice 11 (Pattern: **00001001**)
Contains duplicates
2 → 4 (×2) → 8 (×2) → 16 (×2) → 32 (×2) → 65 (×2+1) → 130 (×2) → 260 (×2) → **521** (×2+1)

The observed branches:

(2, 4, 8**)**
M13, M19, M30, M52
(2, 4, 8, 15**) M13**
(2, 4, 8, 16**) M19**
(2, 4, 8, 16**)** M30, M52
(2, 4, 8, 16, 32**) M4**
(2, 4, 8, 16, 32**)** M30, M52
(2, 4, 8, 16, 32, 64**) M30**
(2, 4, 8, 16, 32, 64**) M52**

How to read this:m **M11** (3, 6, 13, 26, 53**) M11= 107,** Mersennes exponent 107. Each term in black color is multiply by 2 and each term in green color is multiply by 2 and add 1in next term..

So M11= from base 3 x2= 6, multiply by 2 is 12 we add 1 is 13, multiply by 2 is 26 multiply by 2 is 52 we add 1 is 53, we multiply by 2 is 106 we add 1 is 107.

All known Mersennes exponents latices sequences end in green numbers form of nx2+1.

Above is the complete study:

Introduction

Mersenne exponents are prime numbers used in the formula for Mersenne primes, which are of the form 2n−1. The known Mersenne exponents include 2, 3, 5, 7, 13, 17, 19, 31, and others, with the largest known Mersenne prime being $2^{136,279,841} - 1$, as of 2025.

A000043

Mersenne exponents: primes p such that 2^p - 1 is prime. Then 2^p - 1 is called a Mersenne prime.

(Formerly M0672 N0248)

2, 3, 5, 7, 13, 17, 19, 31, 61, 89, 107, 127, 521, 607, 1279, 2203, 2281, 3217, 4253, 4423, 9689, 9941, 11213, 19937, 21701, 23209, 44497, 86243, 110503, 132049, 216091, 756839, 859433, 1257787, 1398269, 2976221, 3021377, 6972593, 13466917, 20996011, 24036583, 25964951, 30402457, 32582657, 37156667, 42643801, 43112609, 57885161, 74207281, 77232917

https://oeis.org/A000043

MERSENNES EXPONENTS SEQUENCES LATTICES in TERMS nx2 and nx2+1

MERSENNES EXPONENTS as SEQUENCES CHAINS LATTICES

LATTICE BUILDER EXPORT
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Export Date: 3/19/2026, 9:56:02 PM
Total numbers used: 579
Lattices created: 50
Duplicates allowed: 50

ALL NUMBERS USED:
2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 49, 50, 51, 52, 53, 55, 57, 61, 62, 63, 64, 65, 66, 68, 69, 70, 71, 73, 75, 76, 77, 78, 79, 80, 81, 82, 84, 85, 86, 87, 89, 90, 91, 92, 99, 100, 102, 104, 105, 106, 107, 110, 115, 124, 127, 128, 129, 130, 132, 137, 138, 141, 142, 147, 151, 153, 155, 157, 159, 160, 162, 164, 168, 169, 170, 173, 175, 181, 183, 184, 198, 201, 205, 209, 211, 212, 215, 220, 231, 248, 257, 259, 260, 265, 275, 276, 283, 285, 294, 302, 303, 307, 310, 311, 315, 319, 320, 325, 328, 336, 339, 341, 347, 350, 362, 363, 366, 368, 369, 396, 402, 410, 419, 422, 425, 431, 441, 463, 497, 515, 519, 521, 531, 550, 552, 566, 570, 589, 605, 607, 614, 621, 623, 630, 639, 640, 650, 657, 673, 678, 682, 695, 700, 725, 726, 733, 737, 739, 792, 804, 821, 839, 844, 851, 863, 883, 927, 994, 1031, 1039, 1063, 1101, 1105, 1132, 1133, 1140, 1178, 1211, 1228, 1242, 1246, 1260, 1279, 1281, 1301, 1315, 1347, 1356, 1365, 1390, 1401, 1450, 1453, 1467, 1475, 1478, 1584, 1608, 1643, 1678, 1688, 1702, 1726, 1766, 1855, 1988, 2063, 2079, 2126, 2203, 2211, 2264, 2267, 2281, 2356, 2422, 2456, 2485, 2492, 2520, 2562, 2602, 2631, 2695, 2712, 2730, 2781, 2803, 2901, 2906, 2934, 2950, 2956, 3169, 3217, 3287, 3357, 3376, 3404, 3453, 3533, 3711, 3977, 4126, 4158, 4253, 4423, 4529, 4535, 4713, 4844, 4913, 4970, 4984, 5040, 5125, 5205, 5262, 5390, 5425, 5461, 5562, 5606, 5802, 5812, 5868, 5901, 5912, 6339, 6575, 6714, 6752, 6809, 6906, 7066, 7422, 7954, 8253, 8317, 9058, 9071, 9427, 9689, 9826, 9941, 9968, 10081, 10251, 10411, 10525, 10780, 10850, 10923, 11124, 11213, 11604, 11625, 11736, 11802, 11825, 12678, 13151, 13428, 13505, 13618, 13812, 14132, 14844, 15909, 16506, 16635, 18117, 18142, 18855, 19652, 19937, 20163, 20503, 20822, 21051, 21560, 21701, 21847, 22248, 23209, 23251, 23473, 23604, 23651, 25356, 26302, 26857, 27011, 27236, 27625, 28264, 29689, 31819, 33012, 33271, 36234, 36285, 37711, 39305, 40327, 41007, 41644, 42102, 43121, 43695, 44497, 46503, 46946, 47209, 47302, 50712, 52605, 53714, 54022, 54473, 55251, 56528, 59379, 63638, 66024, 66542, 72468, 72571, 75422, 78611, 80654, 82015, 83288, 84204, 86243, 87391, 93006, 93892, 94418, 94604, 101425, 105210, 107429, 108045, 108946, 110503, 113056, 118759, 127276, 132049, 133085, 144936, 145143, 150845, 157223, 161308, 164031, 166577, 168408, 174783, 186013, 187785, 188836, 189209, 202851, 210420, 214858, 216091, 217893, 226113, 237519, 254552, 266171, 289872, 290286, 301691, 314446, 322616, 328062, 333154, 336817, 349567, 372027, 375571, 377672, 378419, 405702, 420841, 429716, 435787, 452227, 475038, 509104, 532343, 579744, 580572, 603382, 628893, 645233, 656125, 666309, 673634, 699134, 744055, 751143, 755344, 756839, 811404, 841682, 859433, 871574, 904455, 950076, 1018208, 1064686, 1159488, 1161145, 1206764, 1257787, 1290467, 1312250, 1332618, 1347269, 1398269, 1488110, 1502286, 1510688, 1622809, 1683364, 1743148, 1808911, 1900153, 2036416, 2129372, 2318977, 2322291, 2413528, 2580935, 2624501, 2665237, 2694538, 2976221, 3004572, 3021377, 3245618, 3366729, 3486296, 3617822, 3800307, 4072832, 4258745, 4637955, 4644583, 4827057, 5161870, 5249002, 5330475, 5389076, 6009145, 6491237, 6733458, 6972593, 7235645, 7600614, 8145664, 8517490, 9275910, 9289166, 9654114, 10323741, 10498005, 10660950, 10778152, 12018291, 12982475, 13466917, 14471290, 15201228, 16291328, 17034980, 18551820, 18578333, 19308229, 20647483, 20996011, 21321900, 21556304, 24036583, 25964951, 28942580, 30402457, 32582657, 34069960, 37103640, 37156667, 38616458, 41294966, 42643801, 43112609, 57885161, 68139920, 74207281, 77232917, 82589933, 136279841

NUMBERS WITH DUPLICATES ALLOWED:
4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20, 21, 22, 23, 25, 26, 31, 32, 34, 35, 37, 38, 39, 40, 42, 43, 45, 46, 52, 53, 64, 70, 75, 77, 84, 90, 92, 107, 141, 151, 155, 181, 184, 283, 566

WE HAVE BASE M1= 2 AND BASE M2= 3
PATTERN MEANING: 0 = x 2 AND 1 = x 2 +1

LATTICES:

Lattice 1 (Pattern: 1)
2 → **5** (×2+1)

Lattice 2 (Pattern: 1)
3 → **7** (×2+1)

Lattice 3 (Pattern: 01)
3 → 6 (×2) → **13** (×2+1)

Lattice 4 (Pattern: 001)
2 → 4 (×2) → 8 (×2) → **17** (×2+1)

Lattice 5 (Pattern: 011)
Contains duplicates
2 → 4 (×2) → 9 (×2+1) → **19** (×2+1)

Lattice 6 (Pattern: 111)
Contains duplicates
3 → 7 (×2+1) → 15 (×2+1) → **31** (×2+1)

Lattice 7 (Pattern: 1101)
Contains duplicates
3 → 7 (×2+1) → 15 (×2+1) → 30 (×2) → **61** (×2+1)

Lattice 8 (Pattern: 11001)
Contains duplicates
2 → 5 (×2+1) → 11 (×2+1) → 22 (×2) → 44 (×2) → **89** (×2+1)

Lattice 9 (Pattern: 01011)
Contains duplicates
3 → 6 (×2) → 13 (×2+1) → 26 (×2) → 53 (×2+1) → **107** (×2+1)

Lattice 10 (Pattern: 11111)
Contains duplicates
3 → 7 (×2+1) → 15 (×2+1) → 31 (×2+1) → 63 (×2+1) → **127** (×2+1)

Lattice 11 (Pattern: 00001001)
Contains duplicates
2 → 4 (×2) → 8 (×2) → 16 (×2) → 32 (×2) → 65 (×2+1) → 130 (×2) → 260 (×2) → **521** (×2+1)

Lattice 12 (Pattern: 01011111)
Contains duplicates
2 → 4 (×2) → 9 (×2+1) → 18 (×2) → 37 (×2+1) → 75 (×2+1) → 151 (×2+1) → 303 (×2+1) → **607** (×2+1)

Lattice 13 (Pattern: 011111111)
Contains duplicates
2 → 4 (×2) → 9 (×2+1) → 19 (×2+1) → 39 (×2+1) → 79 (×2+1) → 159 (×2+1) → 319 (×2+1) → 639 (×2+1) → **1279** (×2+1)

Lattice 14 (Pattern: 0010011011)
Contains duplicates
2 → 4 (×2) → 8 (×2) → 17 (×2+1) → 34 (×2) → 68 (×2) → 137 (×2+1) → 275 (×2+1) → 550 (×2) → 1101 (×2+1) → **2203** (×2+1)

Lattice 15 (Pattern: 0011101001)
Contains duplicates
2 → 4 (×2) → 8 (×2) → 17 (×2+1) → 35 (×2+1) → 71 (×2+1) → 142 (×2) → 285 (×2+1) → 570 (×2) → 1140 (×2) → **2281** (×2+1)

Lattice 16 (Pattern: 0010010001)
Contains duplicates
3 → 6 (×2) → 12 (×2) → 25 (×2+1) → 50 (×2) → 100 (×2) → 201 (×2+1) → 402 (×2) → 804 (×2) → 1608 (×2) → **3217** (×2+1)

Lattice 17 (Pattern: 00010011101)

Contains duplicates
2 → 4 (×2) → 8 (×2) → 16 (×2) → 33 (×2+1) → 66 (×2) → 132 (×2) → 265 (×2+1) → 531 (×2+1) → 1063 (×2+1) → 2126 (×2) → **4253** (×2+1)

Lattice 18 (Pattern: 00101000111)
Contains duplicates
2 → 4 (×2) → 8 (×2) → 17 (×2+1) → 34 (×2) → 69 (×2+1) → 138 (×2) → 276 (×2) → 552 (×2) → 1105 (×2+1) → 2211 (×2+1) → **4423** (×2+1)

Lattice 19 (Pattern: **010111011001**)
Contains duplicates
2 → 4 (×2) → 9 (×2+1) → 18 (×2) → 37 (×2+1) → 75 (×2+1) → 151 (×2+1) → 302 (×2) → 605 (×2+1) → 1211 (×2+1) → 2422 (×2) → 4844 (×2) → **9689** (×2+1)

Lattice 20 (Pattern: **011011010101**)
Contains duplicates
2 → 4 (×2) → 9 (×2+1) → 19 (×2+1) → 38 (×2) → 77 (×2+1) → 155 (×2+1) → 310 (×2) → 621 (×2+1) → 1242 (×2) → 2485 (×2+1) → 4970 (×2) → **9941** (×2+1)

Lattice 21 (Pattern: **101111001101**)
Contains duplicates
2 → 5 (×2+1) → 10 (×2) → 21 (×2+1) → 43 (×2+1) → 87 (×2+1) → 175 (×2+1) → 350 (×2) → 700 (×2) → 1401 (×2+1) → 2803 (×2+1) → 5606 (×2) → **11213** (×2+1)

Lattice 22 (Pattern: **0110111100001**)
Contains duplicates
2 → 4 (×2) → 9 (×2+1) → 19 (×2+1) → 38 (×2) → 77 (×2+1) → 155 (×2+1) → 311 (×2+1) → 623 (×2+1) → 1246 (×2) → 2492 (×2) → 4984 (×2) → 9968 (×2) → **19937** (×2+1)

Lattice 23 (Pattern: **1010011000101**)
Contains duplicates
2 → 5 (×2+1) → 10 (×2) → 21 (×2+1) → 42 (×2) → 84 (×2) → 169 (×2+1) → 339 (×2+1) → 678 (×2) → 1356 (×2) → 2712 (×2) → 5425 (×2+1) → 10850 (×2) → **21701** (×2+1)

Lattice 24 (Pattern: **1101010101001**)
Contains duplicates
2 → 5 (×2+1) → 11 (×2+1) → 22 (×2) → 45 (×2+1) → 90 (×2) → 181 (×2+1) → 362 (×2) → 725 (×2+1) → 1450 (×2) → 2901 (×2+1) → 5802 (×2) → 11604 (×2) → **23209** (×2+1)

Lattice 25 (Pattern: **10110111010001**)
Contains duplicates
2 → 5 (×2+1) → 10 (×2) → 21 (×2+1) → 43 (×2+1) → 86 (×2) → 173 (×2+1) → 347 (×2+1) → 695 (×2+1) → 1390 (×2) → 2781 (×2+1) → 5562 (×2) → 11124 (×2) → 22248 (×2) → **44497** (×2+1)

Lattice 26 (Pattern: **101000011100011**)
Contains duplicates
2 → 5 (×2+1) → 10 (×2) → 21 (×2+1) → 42 (×2) → 84 (×2) → 168 (×2) → 336 (×2) → 673 (×2+1) → 1347 (×2+1) → 2695 (×2+1) → 5390 (×2) → 10780 (×2) → 21560 (×2) → 43121 (×2+1) → **86243** (×2+1)

Lattice 27 (Pattern: **010111110100111**)
Contains duplicates
3 → 6 (×2) → 13 (×2+1) → 26 (×2) → 53 (×2+1) → 107 (×2+1) → 215 (×2+1) → 431 (×2+1) → 863 (×2+1) → 1726 (×2) → 3453 (×2+1) → 6906 (×2) → 13812 (×2) → 27625 (×2+1) → 55251 (×2+1) → **110503** (×2+1)

Lattice 28 (Pattern: **0000001111010001**)
Contains duplicates
2 → 4 (×2) → 8 (×2) → 16 (×2) → 32 (×2) → 64 (×2) → 128 (×2) → 257 (×2+1) → 515 (×2+1) → 1031 (×2+1) → 2063 (×2+1) → 4126 (×2) → 8253 (×2+1) → 16506 (×2) → 33012 (×2) → 66024 (×2) → **132049** (×2+1)

Lattice 29 (Pattern: **0100110000011011**)
Contains duplicates
3 → 6 (×2) → 13 (×2+1) → 26 (×2) → 52 (×2) → 105 (×2+1) → 211 (×2+1) → 422 (×2) → 844 (×2) → 1688 (×2) → 3376 (×2) → 6752 (×2) → 13505 (×2+1) → 27011 (×2+1) → 54022 (×2) → 108045 (×2+1) → **216091** (×2+1)

Lattice 30 (Pattern: **111000110001100111**)
Contains duplicates

2 → 5 (×2+1) → 11 (×2+1) → 23 (×2+1) → 46 (×2) → 92 (×2) → 184 (×2) → 369 (×2+1) → 739 (×2+1) → 1478 (×2) → 2956 (×2) → 5912 (×2) → 11825 (×2+1) → 23651 (×2+1) → 47302 (×2) → 94604 (×2) → 189209 (×2+1) → 378419 (×2+1) → **756839** (×2+1)

Lattice 31 (Pattern: 010001110100101001)
Contains duplicates
3 → 6 (×2) → 13 (×2+1) → 26 (×2) → 52 (×2) → 104 (×2) → 209 (×2+1) → 419 (×2+1) → 839 (×2+1) → 1678 (×2) → 3357 (×2+1) → 6714 (×2) → 13428 (×2) → 26857 (×2+1) → 53714 (×2) → 107429 (×2+1) → 214858 (×2) → 429716 (×2) → **859433** (×2+1)

Lattice 32 (Pattern: 01100110001001111011)
Contains duplicates
2 → 4 (×2) → 9 (×2+1) → 19 (×2+1) → 38 (×2) → 76 (×2) → 153 (×2+1) → 307 (×2+1) → 614 (×2) → 1228 (×2) → 2456 (×2) → 4913 (×2+1) → 9826 (×2) → 19652 (×2) → 39305 (×2+1) → 78611 (×2+1) → 157223 (×2+1) → 314446 (×2) → 628893 (×2+1) → **1257787** (×2+1)

Lattice 33 (Pattern: 10101010101111111101)
Contains duplicates
2 → 5 (×2+1) → 10 (×2) → 21 (×2+1) → 42 (×2) → 85 (×2+1) → 170 (×2) → 341 (×2+1) → 682 (×2) → 1365 (×2+1) → 2730 (×2) → 5461 (×2+1) → 10923 (×2+1) → 21847 (×2+1) → 43695 (×2+1) → 87391 (×2+1) → 174783 (×2+1) → 349567 (×2+1) → 699134 (×2) → **1398269** (×2+1)

Lattice 34 (Pattern: 11010110100111011101)
Contains duplicates
2 → 5 (×2+1) → 11 (×2+1) → 22 (×2) → 45 (×2+1) → 90 (×2) → 181 (×2+1) → 363 (×2+1) → 726 (×2) → 1453 (×2+1) → 2906 (×2) → 5812 (×2) → 11625 (×2+1) → 23251 (×2+1) → 46503 (×2+1) → 93006 (×2) → 186013 (×2+1) → 372027 (×2+1) → 744055 (×2+1) → 1488110 (×2) → **2976221** (×2+1)

Lattice 35 (Pattern: 11100001101001000001)
Contains duplicates
2 → 5 (×2+1) → 11 (×2+1) → 23 (×2+1) → 46 (×2) → 92 (×2) → 184 (×2) → 368 (×2) → 737 (×2+1) → 1475 (×2+1) → 2950 (×2) → 5901 (×2+1) → 11802 (×2) → 23604 (×2) → 47209 (×2+1) → 94418 (×2) → 188836 (×2) → 377672 (×2) → 755344 (×2) → 1510688 (×2) → **3021377** (×2+1)

Lattice 36 (Pattern: 01010011001001011001 1)
Contains duplicates
3 → 6 (×2) → 13 (×2+1) → 26 (×2) → 53 (×2+1) → 106 (×2) → 212 (×2) → 425 (×2+1) → 851 (×2+1) → 1702 (×2) → 3404 (×2) → 6809 (×2+1) → 13618 (×2) → 27236 (×2) → 54473 (×2+1) → 108946 (×2) → 217893 (×2+1) → 435787 (×2+1) → 871574 (×2) → 1743148 (×2) → 3486296 (×2) → **6972593** (×2+1)

Lattice 37 (Pattern: 0011010111110100100101)
Contains duplicates
3 → 6 (×2) → 12 (×2) → 25 (×2+1) → 51 (×2+1) → 102 (×2) → 205 (×2+1) → 410 (×2) → 821 (×2+1) → 1643 (×2+1) → 3287 (×2+1) → 6575 (×2+1) → 13151 (×2+1) → 26302 (×2) → 52605 (×2+1) → 105210 (×2) → 210420 (×2) → 420841 (×2+1) → 841682 (×2) → 1683364 (×2) → 3366729 (×2+1) → 6733458 (×2) → **13466917** (×2+1)

Lattice 38 (Pattern: 100000001011111110101011)
Contains duplicates
2 → 5 (×2+1) → 10 (×2) → 20 (×2) → 40 (×2) → 80 (×2) → 160 (×2) → 320 (×2) → 640 (×2) → 1281 (×2+1) → 2562 (×2) → 5125 (×2+1) → 10251 (×2+1) → 20503 (×2+1) → 41007 (×2+1) → 82015 (×2+1) → 164031 (×2+1) → 328062 (×2) → 656125 (×2+1) → 1312250 (×2) → 2624501 (×2+1) → 5249002 (×2) → 10498005 (×2+1) → **20996011** (×2+1)

Lattice 39 (Pattern: 11011101100010011100111)
Contains duplicates
2 → 5 (×2+1) → 11 (×2+1) → 22 (×2) → 45 (×2+1) → 91 (×2+1) → 183 (×2+1) → 366 (×2) → 733 (×2+1) → 1467 (×2+1) → 2934 (×2) → 5868 (×2) → 11736 (×2) → 23473 (×2+1) → 46946 (×2) → 93892 (×2) → 187785 (×2+1) → 375571 (×2+1) → 751143 (×2+1) → 1502286 (×2) → 3004572 (×2) → 6009145 (×2+1) → 12018291 (×2+1) → **24036583** (×2+1)

Lattice 40 (Pattern: 0001100001100011001011 1)
Contains duplicates
3 → 6 (×2) → 12 (×2) → 24 (×2) → 49 (×2+1) → 99 (×2+1) → 198 (×2) → 396 (×2) → 792 (×2) → 1584 (×2) → 3169 (×2+1) → 6339 (×2+1) → 12678 (×2) → 25356 (×2) → 50712 (×2) → 101425 (×2+1) → 202851 (×2+1) → 405702 (×2) → 811404 (×2) → 1622809 (×2+1) → 3245618 (×2) → 6491237 (×2+1) → 12982475 (×2+1) → **25964951** (×2+1)

Lattice 41 (Pattern: 1001111111001111001 1001)
Contains duplicates
3 → 7 (×2+1) → 14 (×2) → 28 (×2) → 57 (×2+1) → 115 (×2+1) → 231 (×2+1) → 463 (×2+1) → 927 (×2+1) → 1855 (×2+1) → 3711 (×2+1) → 7422 (×2) → 14844 (×2) → 29689 (×2+1) → 59379 (×2+1) → 118759 (×2+1) → 237519 (×2+1) → 475038 (×2) → 950076 (×2) → 1900153 (×2+1) → 3800307 (×2+1) → 7600614 (×2) → 15201228 (×2) → **30402457** (×2+1)

Lattice 42 (Pattern: 11100010010110000000001)

Contains duplicates
3 → 7 (×2+1) → 15 (×2+1) → 31 (×2+1) → 62 (×2) → 124 (×2) → 248 (×2) → 497 (×2+1) → 994 (×2) → 1988 (×2) → 3977 (×2+1) → 7954 (×2) → 15909 (×2+1) → 31819 (×2+1) → 63638 (×2) → 127276 (×2) → 254552 (×2) → 509104 (×2) → 1018208 (×2) → 2036416 (×2) → 4072832 (×2) → 8145664 (×2) → 16291328 (×2) → **32582657** (×2+1)

Lattice 43 (Pattern: 001101101111011100111011)
Contains duplicates
2 → 4 (×2) → 8 (×2) → 17 (×2+1) → 35 (×2+1) → 70 (×2) → 141 (×2+1) → 283 (×2+1) → 566 (×2) → 1133 (×2+1) → 2267 (×2+1) → 4535 (×2+1) → 9071 (×2+1) → 18142 (×2) → 36285 (×2+1) → 72571 (×2+1) → 145143 (×2+1) → 290286 (×2) → 580572 (×2) → 1161145 (×2+1) → 2322291 (×2+1) → 4644583 (×2+1) → 9289166 (×2) → 18578333 (×2+1) → **37156667** (×2+1)

Lattice 44 (Pattern: 100010101011000101011001)
Contains duplicates
2 → 5 (×2+1) → 10 (×2) → 20 (×2) → 40 (×2) → 81 (×2+1) → 162 (×2) → 325 (×2+1) → 650 (×2) → 1301 (×2+1) → 2602 (×2) → 5205 (×2+1) → 10411 (×2+1) → 20822 (×2) → 41644 (×2) → 83288 (×2) → 166577 (×2+1) → 333154 (×2) → 666309 (×2+1) → 1332618 (×2) → 2665237 (×2+1) → 5330475 (×2+1) → 10660950 (×2) → 21321900 (×2) → **42643801** (×2+1)

Lattice 45 (Pattern: 100100011101100010100001)
Contains duplicates
2 → 5 (×2+1) → 10 (×2) → 20 (×2) → 41 (×2+1) → 82 (×2) → 164 (×2) → 328 (×2) → 657 (×2+1) → 1315 (×2+1) → 2631 (×2+1) → 5262 (×2) → 10525 (×2+1) → 21051 (×2+1) → 42102 (×2) → 84204 (×2) → 168408 (×2) → 336817 (×2+1) → 673634 (×2) → 1347269 (×2+1) → 2694538 (×2) → 5389076 (×2) → 10778152 (×2) → 21556304 (×2) → **43112609** (×2+1)

Lattice 46 (Pattern: 011100110100000111101001)
Contains duplicates
3 → 6 (×2) → 13 (×2+1) → 27 (×2+1) → 55 (×2+1) → 110 (×2) → 220 (×2) → 441 (×2+1) → 883 (×2+1) → 1766 (×2) → 3533 (×2+1) → 7066 (×2) → 14132 (×2) → 28264 (×2) → 56528 (×2) → 113056 (×2) → 226113 (×2+1) → 452227 (×2+1) → 904455 (×2+1) → 1808911 (×2+1) → 3617822 (×2) → 7235645 (×2+1) → 14471290 (×2) → 28942580 (×2) → **57885161** (×2+1)

Lattice 47 (Pattern: 001101100010100000110001)
Contains duplicates
2 → 4 (×2) → 8 (×2) → 17 (×2+1) → 35 (×2+1) → 70 (×2) → 141 (×2+1) → 283 (×2+1) → 566 (×2) → 1132 (×2) → 2264 (×2) → 4529 (×2+1) → 9058 (×2) → 18117 (×2+1) → 36234 (×2) → 72468 (×2) → 144936 (×2) → 289872 (×2) → 579744 (×2) → 1159488 (×2) → 2318977 (×2+1) → 4637955 (×2+1) → 9275910 (×2) → 18551820 (×2) → 37103640 (×2) → **74207281** (×2+1)

Lattice 48 (Pattern: 010011010011110110001010 1)
Contains duplicates
2 → 4 (×2) → 9 (×2+1) → 18 (×2) → 36 (×2) → 73 (×2+1) → 147 (×2+1) → 294 (×2) → 589 (×2+1) → 1178 (×2) → 2356 (×2) → 4713 (×2+1) → 9427 (×2+1) → 18855 (×2+1) → 37711 (×2+1) → 75422 (×2) → 150845 (×2+1) → 301691 (×2+1) → 603382 (×2) → 1206764 (×2) → 2413528 (×2) → 4827057 (×2+1) → 9654114 (×2) → 19308229 (×2+1) → 38616458 (×2) → **77232917** (×2+1)

Lattice 49 (Pattern: 011101100001110001110110 1)
Contains duplicates
2 → 4 (×2) → 9 (×2+1) → 19 (×2+1) → 39 (×2+1) → 78 (×2) → 157 (×2+1) → 315 (×2+1) → 630 (×2) → 1260 (×2) → 2520 (×2) → 5040 (×2) → 10081 (×2+1) → 20163 (×2+1) → 40327 (×2+1) → 80654 (×2) → 161308 (×2) → 322616 (×2) → 645233 (×2+1) → 1290467 (×2+1) → 2580935 (×2+1) → 5161870 (×2) → 10323741 (×2+1) → 20647483 (×2+1) → 41294966 (×2) → **82589933** (×2+1)

Lattice 50 (Pattern: 000001111101110111001000 01)
Contains duplicates
2 → 4 (×2) → 8 (×2) → 16 (×2) → 32 (×2) → 64 (×2) → 129 (×2+1) → 259 (×2+1) → 519 (×2+1) → 1039 (×2+1) → 2079 (×2+1) → 4158 (×2) → 8317 (×2+1) → 16635 (×2+1) → 33271 (×2+1) → 66542 (×2) → 133085 (×2+1) → 266171 (×2+1) → 532343 (×2+1) → 1064686 (×2) → 2129372 (×2) → 4258745 (×2+1) → 8517490 (×2) → 17034980 (×2) → 34069960 (×2) → 68139920 (×2) → **136279841** (×2+1)

The duplicate numbers are allowing the grow and branching (numbers is red are duplicate numbers).

M1 = 2 - BRANCHES

Lattice 1 pattern: 1 **1 LEG**

2 → **5**(×2+1)

Lattice 2 pattern: 001 **1 LEG 2 SHELLS**

2 → 4×2 → 8×2 → **17**(×2+1)

Lattice 3 (has duplicates) pattern: 0010011011 **5 LEGS 5 SHELLS**

2 → 4×2 → 8×2 → 17(×2+1) → 34×2 → 68×2 → 137(×2+1) → 275(×2+1) → 550×2 → 1101(×2+1) → **2203**(×2+1)

Lattice 4 (has duplicates) pattern: **0011101001** **5 LEGS 5 SHELLS**

2 → **4**×2 → **8**×2 → **17**(×2+1) → 35(×2+1) → 71(×2+1) → 142×2 → 285(×2+1) → 570×2 → 1140×2 → **2281**(×2+1)

Lattice 5 (has duplicates) pattern: **00101000111** **5 LEGS 6 SHELLS**

2 → **4**×2 → **8**×2 → **17**(×2+1) → **34**×2 → 69(×2+1) → 138×2 → 276×2 → 552×2 → 1105(×2+1) → 2211(×2+1) → **4423**(×2+1)

Lattice 6 (has duplicates) pattern: **001101101111011100111011** **16 LEGS 8 SHELLS**

2 → **4**×2 → **8**×2 → **17**(×2+1) → **35**(×2+1) → 70×2 → 141(×2+1) → 283(×2+1) → 566×2 → 1133(×2+1) → 2267(×2+1) → 4535(×2+1) → 9071(×2+1) → 18142×2 → 36285(×2+1) → 72571(×2+1) → 145143(×2+1) → 290286×2 → 580572×2 → 1161145(×2+1) → 2322291(×2+1) → 4644583(×2+1) → 9289166×2 → 18578333(×2+1) → **37156667**(×2+1)

Lattice 7 (has duplicates) pattern: **001101100010100000011000 01** **9 LEGS 16 SHELLS**

2 → **4**×2 → **8**×2 → **17**(×2+1) → **35**(×2+1) → **70**×2 → **141**(×2+1) → **283**(×2+1) → **566**×2 → 1132×2 → 2264×2 → 4529(×2+1) → 9058×2 → 18117(×2+1) → 36234×2 → 72468×2 → 144936×2 → 289872×2 → 579744×2 → 1159488×2 → 2318977(×2+1) → 4637955(×2+1) → 9275910×2 → 18551820×2 → 37103640×2 → **74207281**(×2+1)

Lattice 1 pattern: **1** **1 LEG**

2 → **5**(×2+1)

Lattice 2 pattern: **011** **2 LEGS 1 SHELL**

2 → 4×2 → 9(×2+1) → **19**(×2+1)

Lattice 3 (has duplicates) pattern: **011011010101** **7 LEGS 5 SHELLS**

2 → **4**×2 → **9**(×2+1) → **19**(×2+1) → 38×2 → 77(×2+1) → 155(×2+1) → 310×2 → 621(×2+1) → 1242×2 → 2485(×2+1) → 4970×2 → **9941**(×2+1)

Lattice 4 (has duplicates) pattern: **0110111100001** **7 LEGS 6 SHELLS**

2 → **4**×2 → **9**(×2+1) → **19**(×2+1) → **38**×2 → **77**(×2+1) → **155**(×2+1) → 311(×2+1) → 623(×2+1) → 1246×2 → 2492×2 → 4984×2 → 9968×2 → **19937**(×2+1)

Lattice 5 (has duplicates) pattern: **0110011000100111011** **10 LEGS 9 SHELLS**

2 → **4**×2 → **9**(×2+1) → **19**(×2+1) → **38**×2 → 76×2 → 153(×2+1) → 307(×2+1) → 614×2 → 1228×2 → 2456×2 → 4913(×2+1) → 9826×2 → 19652×2 → 39305(×2+1) → 78611(×2+1) → 157223(×2+1) → 314446×2 → 628893(×2+1) → **1257787**(×2+1)

Lattice 6 (has duplicates) pattern: **0111011000011100011101101** **14 LEGS 11 SHELLS**

2 → **4**×2 → **9**(×2+1) → **19**(×2+1) → 39(×2+1) → 78×2 → 157(×2+1) → 315(×2+1) → 630×2 → 1260×2 → 2520×2 → 5040×2 → 10081(×2+1) → 20163(×2+1) → 40327(×2+1) → 80654×2 → 161308×2 → 322616×2 → 645233(×2+1) → 1290467(×2+1) → 2580935(×2+1) → 5161870×2 → 10323741(×2+1) → 20647483(×2+1) → 41294966×2 → **82589933**(×2+1)

Lattice 1 pattern: **1** **1 LEG**

2 → **5**(×2+1)

Lattice 2 (has duplicates) pattern: **11001** **3 LEGS 2 SHELLS**

2 → **5**(×2+1) → 11(×2+1) → 22×2 → 44×2 → **89**(×2+1)

Lattice 3 (has duplicates) pattern: **1101010101001** **7 LEGS 6 SHELLS**

2 → **5**(×2+1) → **11**(×2+1) → **22**×2 → 45(×2+1) → 90×2 → 181(×2+1) → 362×2 → 725(×2+1) → 1450×2 → 2901(×2+1) → 5802×2 → 11604×2 → **23209**(×2+1)

Lattice 4 (has duplicates) pattern: **11010110100111011101** **13 LEGS 7 SHELLS**

2 → **5**(×2+1) → **11**(×2+1) → **22**×2 → **45**(×2+1) → **90**×2 → **181**(×2+1) → 363(×2+1) → 726×2 → 1453(×2+1) → 2906×2 → 5812×2 → 11625(×2+1) → 23251(×2+1) → 46503(×2+1) → 93006×2 → 186013(×2+1) → 372027(×2+1) → 744055(×2+1) → 1488110×2 → **2976221**(×2+1)

Lattice 5 (has duplicates) pattern: **11011101100010011100111** **14 LEGS 9 SHELLS**

2 → **5**(×2+1) → **11**(×2+1) → **22**×2 → **45**(×2+1) → 91(×2+1) → 183(×2+1) → 366×2 → 733(×2+1) → 1467(×2+1) → 2934×2 → 5868×2 → 11736×2 → 23473(×2+1) → 46946×2 → 93892×2 → 187785(×2+1) → 375571(×2+1) → 751143(×2+1) → 1502286×2 → 3004572×2 → 6009145(×2+1) → 12018291(×2+1) → **24036583**(×2+1)

Lattice 1 pattern: **1** **1 LEG**

2 → **5**(×2+1)

Lattice 2 pattern: **00001001** **2 LEGS 6 SHELLS**

2 → 4×2 → 8×2 → 16×2 → 32×2 → 65(×2+1) → 130×2 → 260×2 → **521**(×2+1)

Lattice 3 (has duplicates) pattern: **00010011101** **5 LEGS 6 SHELLS**

2 → **4**×2 → **8**×2 → **16**×2 → 33(×2+1) → 66×2 → 132×2 → 265(×2+1) → 531(×2+1) → 1063(×2+1) → 2126×2 → **4253**(×2+1)

Lattice 4 (has duplicates) pattern: **0000001111010001** **6 LEGS 10 SHELLS**

2 → **4**×2 → **8**×2 → **16**×2 → **32**×2 → 64×2 → 128×2 → 257(×2+1) → 515(×2+1) → 1031(×2+1) → 2063(×2+1) → 4126×2 → 8253(×2+1) → 16506×2 → 33012×2 → 66024×2 → **132049**(×2+1)

Lattice 5 (has duplicates) pattern: **0000011111011011011100100001** **13 LEGS 13 SHELLS**

2 → **4**×2 → **8**×2 → **16**×2 → **32**×2 → **64**×2 → 129(×2+1) → 259(×2+1) → 519(×2+1) → 1039(×2+1) → 2079(×2+1) → 4158×2 → 8317(×2+1) → 16635(×2+1) → 33271(×2+1) → 66542×2 → 133085(×2+1) → 266171(×2+1) → 532343(×2+1) → 1064686×2 → 2129372×2 → 4258745(×2+1) → 8517490×2 → 17034980×2 → 34069960×2 → 68139920×2 → **136279841**(×2+1)

Lattice 1 pattern: **1** **1 LEG**

2 → **5**(×2+1)

Lattice 2 pattern: **01011111** **6 LEGS 2 SHELLS**

2 → 4×2 → 9(×2+1) → 18×2 → 37(×2+1) → 75(×2+1) → 151(×2+1) → 303(×2+1) → **607**(×2+1)

Lattice 3 (has duplicates) pattern: **010111011001** **7 LEGS 5 SHELLS**

2 → **4**×2 → **9**(×2+1) → **18**×2 → **37**(×2+1) → **75**(×2+1) → **151**(×2+1) → 302×2 → 605(×2+1) → 1211(×2+1) → 2422×2 → 4844×2 → **9689**(×2+1)

Lattice 4 (has duplicates) pattern: **0100110100111101100010101** **13 LEGS 12 SHELLS**

2 → **4**×2 → **9**(×2+1) → **18**×2 → 36×2 → 73(×2+1) → 147(×2+1) → 294×2 → 589(×2+1) → 1178×2 → 2356×2 → 4713(×2+1) → 9427(×2+1) → 18855(×2+1) → 37711(×2+1) → 75422×2 → 150845(×2+1) → 301691(×2+1) → 603382×2 → 1206764×2 → 2413528×2 → 4827057(×2+1) → 9654114×2 → 19308229(×2+1) → 38616458×2 → **77232917**(×2+1)

Lattice 1 pattern: **1** **1 LEG**

2 → **5**(×2+1)

Lattice 2 pattern: **011111111** **8 LEGS 1 SHELL**

2 → 4×2 → 9(×2+1) → 19(×2+1) → 39(×2+1) → 79(×2+1) → 159(×2+1) → 319(×2+1) → 639(×2+1) → **1279**(×2+1)

Lattice 1 pattern: **1** **1 LEG**

2 → **5**(×2+1)

Lattice 2 (has duplicates) pattern: **101111001101** **8 LEGS 4 SHELLS**

2 → **5**(×2+1) → 10×2 → 21(×2+1) → 43(×2+1) → 87(×2+1) → 175(×2+1) → 350×2 → 700×2 → 1401(×2+1) → 2803(×2+1) → 5606×2 → **11213**(×2+1)

Lattice 3 (has duplicates) pattern: **1010011000101** **6 LEGS 7 SHELLS**

2 → **5**(×2+1) → **10**×2 → **21**(×2+1) → 42×2 → 84×2 → 169(×2+1) → 339(×2+1) → 678×2 → 1356×2 → 2712×2 → 5425(×2+1) → 10850×2 → **21701**(×2+1)

Lattice 4 (has duplicates) pattern: **10110111010001** **8 LEGS 6 SHELLS**

2 → **5**(×2+1) → **10**×2 → **21**(×2+1) → **43**(×2+1) → 86×2 → 173(×2+1) → 347(×2+1) → 695(×2+1) → 1390×2 → 2781(×2+1) → 5562×2 → 11124×2 → 22248×2 → **44497**(×2+1)

Lattice 5 (has duplicates) pattern: **101000011100011** **7 LEGS 8 SHELLS**

2 → **5**(×2+1) → **10**×2 → **21**(×2+1) → **42**×2 → **84**×2 → 168×2 → 336×2 → 673(×2+1) → 1347(×2+1) → 2695(×2+1) → 5390×2 → 10780×2 → 21560×2 → 43121(×2+1) → **86243**(×2+1)

Lattice 6 (has duplicates) pattern: **1010101010111111101** **13 LEGS 6 SHELLS**

2 → **5**(×2+1) → **10**×2 → **21**(×2+1) → **42**×2 → 85(×2+1) → 170×2 → 341(×2+1) → 682×2 → 1365(×2+1) → 2730×2 → 5461(×2+1) → 10923(×2+1) → 21847(×2+1) → 43695(×2+1) → 87391(×2+1) → 174783(×2+1) → 349567(×2+1) → 699134×2 → **1398269**(×2+1)

Lattice 1 pattern: **1** **1 LEG**

2 → **5**(×2+1)

Lattice 2 (has duplicates) pattern: **111000110001100111** **10 LEGS 8 SHELLS**

2 → **5**(×2+1) → 11(×2+1) → 23(×2+1) → 46×2 → 92×2 → 184×2 → 369(×2+1) → 739(×2+1) → 1478×2 → 2956×2 → 5912×2 → 11825(×2+1) → 23651(×2+1) → 47302×2 → 94604×2 → 189209(×2+1) → 378419(×2+1) → **756839**(×2+1)

Lattice 3 (has duplicates) pattern: **11100001101001000001** **8 LEGS 12 SHELLS**

2 → **5**(×2+1) → **11**(×2+1) → **23**(×2+1) → **46**×2 → **92**×2 → **184**×2 → 368×2 → 737(×2+1) → 1475(×2+1) → 2950×2 → 5901(×2+1) → 11802×2 → 23604×2 → 47209(×2+1) → 94418×2 → 188836×2 → 377672×2 → 755344×2 → 1510688×2 → **3021377**(×2+1)

Lattice 1 pattern: **1** **1 LEG**

2 → **5**(×2+1)

Lattice 2 (has duplicates) pattern: **1000000010111110101011** **12 LEGS 11 SHELLS**

2 → **5**(×2+1) → 10×2 → 20×2 → 40×2 → 80×2 → 160×2 → 320×2 → 640×2 → 1281(×2+1) → 2562×2 → 5125(×2+1) → 10251(×2+1) → 20503(×2+1) → 41007(×2+1) → 82015(×2+1) → 164031(×2+1) → 328062×2 → 656125(×2+1) → 1312250×2 → 2624501(×2+1) → 5249002×2 → 10498005(×2+1) → **20996011**(×2+1)

Lattice 3 (has duplicates) pattern: **1000101010110001010110001** **11 LEGS 13 SHELLS**

2 → **5**(×2+1) → **10**×2 → **20**×2 → **40**×2 → 81(×2+1) → 162×2 → 325(×2+1) → 650×2 → 1301(×2+1) → 2602×2 → 5205(×2+1) → 10411(×2+1) → 20822×2 → 41644×2 → 83288×2 → 166577(×2+1) → 333154×2 → 666309(×2+1) → 1332618×2 → 2665237(×2+1) → 5330475(×2+1) → 10660950×2 → 21321900×2 → **42643801**(×2+1)

Lattice 4 (has duplicates) pattern: **1001000111011000101000001** **10 LEGS 14 SHELLS**

2 → **5**(×2+1) → **10**×2 → **20**×2 → 41(×2+1) → 82×2 → 164×2 → 328×2 → 657(×2+1) → 1315(×2+1) → 2631(×2+1) → 5262×2 → 10525(×2+1) → 21051(×2+1) → 42102×2 → 84204×2 → 168408×2 → 336817(×2+1) → 673634×2 → 1347269(×2+1) → 2694538×2 → 5389076×2 → 10778152×2 → 21556304×2 → **43112609**(×2+1)

M2 = 3 - BRANCHES

Lattice 1 pattern: 1 1 LEG

3 → **7**(×2+1)

Lattice 2 pattern: **01** **1 LEG 1 SHELL**

3 → 6×2 → **13**(×2+1)

Lattice 3 (has duplicates) pattern: **01011** **3 LEGS 2 SHELLS**

3 → **6**×2 → **13**(×2+1) → 26×2 → 53(×2+1) → **107**(×2+1)

Lattice 4 (has duplicates) pattern: 010111110100111 **10 LEGS 5 SHELLS**

3 → 6×2 → 13(×2+1) → 26×2 → 53(×2+1) → 107(×2+1) → 215(×2+1) → 431(×2+1) → 863(×2+1) → 1726×2 → 3453(×2+1) → 6906×2 → 13812×2 → 27625(×2+1) → 55251(×2+1) → **110503**(×2+1)

Lattice 5 (has duplicates) pattern: 010100110010010110001 **9 LEGS 12 SHELLS**

3 → 6×2 → 13(×2+1) → 26×2 → 53(×2+1) → 106×2 → 212×2 → 425(×2+1) → 851(×2+1) → 1702×2 → 3404×2 → 6809(×2+1) → 13618×2 → 27236×2 → 54473(×2+1) → 108946×2 → 217893(×2+1) → 435787(×2+1) → 871574×2 → 1743148×2 → 3486296×2 → **6972593**(×2+1)

Lattice 6 (has duplicates) pattern: 0100110000011011 **7 LEGS 9 SHELLS**

3 → 6×2 → 13(×2+1) → 26×2 → 52×2 → 105(×2+1) → 211(×2+1) → 422×2 → 844×2 → 1688×2 → 3376×2 → 6752×2 → 13505(×2+1) → 27011(×2+1) → 54022×2 → 108045(×2+1) → **216091**(×2+1)

Lattice 7 (has duplicates) pattern: 010001110100101001 **8 LEGS 10 SHELLS**

3 → 6×2 → 13(×2+1) → 26×2 → 52×2 → 104×2 → 209(×2+1) → 419(×2+1) → 839(×2+1) → 1678×2 → 3357(×2+1) → 6714×2 → 13428×2 → 26857(×2+1) → 53714×2 → 107429(×2+1) → 214858×2 → 429716×2 → **859433**(×2+1)

Lattice 1 pattern: 1 1 LEG

3 → **7**(×2+1)

Lattice 2 (has duplicates) pattern: 111 **3 LEGS**

3 → 7(×2+1) → 15(×2+1) → **31**(×2+1)

Lattice 3 (has duplicates) pattern: 1101 **3 LEGS 1 SHELL**

3 → 7(×2+1) → 15(×2+1) → 30×2 → **61**(×2+1)

Lattice 4 (has duplicates) pattern: 11111 **5 LEGS**

3 → 7(×2+1) → 15(×2+1) → 31(×2+1) → 63(×2+1) → **127**(×2+1)

Lattice 5 (has duplicates) pattern: 11100010010110000000001 **8 LEGS 15 SHELLS**

3 → 7(×2+1) → 15(×2+1) → 31(×2+1) → 62×2 → 124×2 → 248×2 → 497(×2+1) → 994×2 → 1988×2 → 3977(×2+1) → 7954×2 → 15909(×2+1) → 31819(×2+1) → 63638×2 → 127276×2 → 254552×2 → 509104×2 → 1018208×2 → 2036416×2 → 4072832×2 → 8145664×2 → 16291328×2 → **32582657**(×2+1)

Lattice 1 pattern: 1 1 LEG

3 → **7**(×2+1)

Lattice 2 pattern: 0010010001 **3 LEGS 7 SHELLS**

3 → 6×2 → 12×2 → 25(×2+1) → 50×2 → 100×2 → 201(×2+1) → 402×2 → 804×2 → 1608×2 → **3217**(×2+1)

Lattice 3 (has duplicates) pattern: 0011010111110100100101 **12 LEGS 10 SHELLS**

3 → 6×2 → 12×2 → 25(×2+1) → 51(×2+1) → 102×2 → 205(×2+1) → 410×2 → 821(×2+1) → 1643(×2+1) → 3287(×2+1) → 6575(×2+1) → 13151(×2+1) → 26302×2 → 52605(×2+1) → 105210×2 → 210420×2 → 420841(×2+1) → 841682×2 → 1683364×2 → 3366729(×2+1) → 6733458×2 → **13466917**(×2+1)

Lattice 4 (has duplicates) pattern: 000110000110001100010111 **10 LEGS 13 SHELLS**

3 → 6×2 → 12×2 → 24×2 → 49(×2+1) → 99(×2+1) → 198×2 → 396×2 → 792×2 → 1584×2 → 3169(×2+1) → 6339(×2+1) → 12678×2 → 25356×2 → 50712×2 → 101425(×2+1) → 202851(×2+1) → 405702×2 → 811404×2 → 1622809(×2+1) → 3245618×2 → 6491237(×2+1) → 12982475(×2+1) → **25964951**(×2+1)

Lattice 1 pattern: 1 1 LEG

3 → **7**(×2+1)

Lattice 2 (has duplicates) pattern: 100111111100111100011001 **15 LEGS 8 SHELLS**

3 → 7(×2+1) → 14×2 → 28×2 → 57(×2+1) → 115(×2+1) → 231(×2+1) → 463(×2+1) → 927(×2+1) → 1855(×2+1) → 3711(×2+1) → 7422×2 → 14844×2 → 29689(×2+1) → 59379(×2+1) → 118759(×2+1) → 237519(×2+1) → 475038×2 → 950076×2 → 1900153(×2+1) → 3800307(×2+1) → 7600614×2 → 15201228×2 → **30402457**(×2+1)

Lattice 1 pattern: 1 1 LEG

3 → **7**(×2+1)

Lattice 2 pattern: **011100110100000111101001 12 LEGS 12 SHELLS**

3 → 6×2 → 13(×2+1) → 27(×2+1) → 55(×2+1) → 110×2 → 220×2 → 441(×2+1) → 883(×2+1) → 1766×2 → 3533(×2+1) → 7066×2 → 14132×2 → 28264×2 → 56528×2 → 113056×2 → 226113(×2+1) → 452227(×2+1) → 904455(×2+1) → 1808911(×2+1) → 3617822×2 → 7235645(×2+1) → 14471290×2 → 28942580×2 → **57885161**(×2+1)

Observation:

Patterns observed by terms:

L= 2n +1

S= 2n

M1, 2= base

M2, 3 =base

M3, 5 (1L) 1 term

2

1

M4, 7 (1L) 1 term

3

1

M5, 13 (1L -1S) 2 terms

3, 6

01

M6, 17 (1L-2S) 3 terms

2, 4, 8

001

M7, 19 (2L-1S) 3 terms

2, 4, 9

011

M8, 31 (3L) 3 terms

3, 7, 15

111

M9, 61 (3L-1S) 4 terms

3, 7, 15, 30

1101

M10, 89 (3L-2S) 5 terms

2, 5, 11, 22, 44

11001

M11, 107 (3L-2S) 5 terms

3, 6, 13, 26, 53

01011

M12, 127 (5L) 5 terms

3, 7, 15, 31, 63

11111

M13, 521 (2L-6S) 8 terms

2, 4, 8, 16, 32, 65, 130, 260

00001001

M14, 607 (6L-2S) 8 terms
2, 4, 9, 18, 37, 75, 151, 303
01011111
M15, 1279 (8L-1S) 9 terms
2, 4, 9, 19, 39, 79, 159, 319, 639
011111111
M16, 2203 (5L-5S) 10 terms
2, 4, 8, 17, 34, 68, 137, 275, 551, 1101
0010011011
M17, 2281 (5L-5S) 10 terms
2, 4, 8, 17, 35, 71, 142, 285, 570, 1140
0011101001
M18, 3217 (3L-7S) 10 terms
3, 6, 12, 25, 50, 100, 201, 402, 804, 1608
0010010001
M19, 4253 (5L-6S) 11 terms
2, 4, 8, 16, 33, 66, 132, 265, 531, 1063, 2126
00010011101
M20, 4423 (5L-6S) 11 terms
2, 4, 8, 17, 34, 69, 138, 276, 552, 1105, 2211
00101000111
M21, 9689 (7L-5S) 12 terms
2, 4, 9, 18, 37, 75, 151, 302, 605, 1211, 2422, 4844
010111011001
M22, 9941 (7L-5S) 12 terms
2, 4, 9, 19, 38, 77, 155, 310, 621, 1242, 2485, 4970
011011010101
M23, 11213 (8L-4S) 12 terms
2, 5, 10, 21, 43, 87, 175, 350, 700, 1401, 2803, 5606
101111001101
M24, 19937 (7L-6S) 13 terms
2, 4, 9, 19, 38, 77, 155, 311, 623, 1246, 2492, 4984, 9968
0110111100001
M25, 21701 (6L-7S) 13 terms
2, 5, 10, 21, 42, 84, 169, 339, 678, 1356, 2712, 5425, 10850
1010011000101
M26, 23209 (7L-6S) 13 terms
2, 5, 11, 22, 45, 90, 181, 362, 725, 1450, 2901, 5802, 11604
1101010101001
M27, 44497 (8L-6S) 14 terms
2, 5, 10, 21, 43, 86, 173, 347, 695, 1390, 2781, 5562, 11124, 22248
10110111010001
M28, 86243 (7L-8S) 15 terms
2, 5, 10, 21, 42, 84, 168, 336, 673, 1347, 2695, 5390, 10780, 21560, 43121
101000011100011
M29, 110503 (10L-5S) 15 terms
3, 6, 13, 26, 53, 107, 215, 431, 863, 1726, 3453, 6906, 13812, 27625, 55251
010111110100111
M30, 132049 (6L-10S) 16 terms
2, 4, 8, 16, 32, 64, 128, 257, 515, 1031, 2063, 4126, 8253, 16506, 33012, 66024

0000001111010001
M31, 216091 (7L-9S) 16 terms
3, 6, 13, 26, 52, 105, 211, 422, 844, 1688, 3376, 6752, 13505, 27011, 54022, 108045
0100110000011011
M32, 756839 (10L-8S) 18 terms
2, 5, 11, 23, 46, 92, 184, 369, 739, 1478, 2956, 5912, 11825, 23651, 47302, 94604, 189209, 378419
111000110001100111
M33, 859433 (8L-10S) 18 terms
3, 6, 13, 26, 52, 104, 209, 419, 839, 1678, 3357, 6714, 13428, 26857, 53714, 107429, 214858, 429716
010001110100101001
M34, 1257787 (10L-9S) 19 terms
2, 4, 9, 19, 38, 76, 153, 307, 614, 1228, 2456, 4913, 9826, 19652, 39305, 78611, 157223, 314446, 628893
0110011000100111011
M35, 1398269 (13L-6S) 19 terms
2, 5, 10, 21, 42, 85, 170, 341, 682, 1365, 2730, 5461, 10923, 21847, 43695, 87391, 174783, 349567, 699134
1010101010111111101
M36, 2976221 (13L-7S) 20 terms
2, 5, 11, 22, 45, 90, 181, 363, 726, 1453, 2906, 5812, 11625, 23251, 46503, 93006, 186013, 372027, 744055, 1488110
11010110100111011101
M37, 3021377 (8L-12S) 20 terms
2, 5, 11, 23, 46, 92, 184, 368, 737, 1475, 2950, 5901, 11802, 23604, 47209, 94418, 188836, 377672, 755344, 1510688
11100001101001000001
M38, 6972593 (9L-12S) 21 terms
3, 6, 13, 26, 53, 106, 212, 425, 851, 1702, 3404, 6809, 13618, 27236, 54473, 108946, 217893, 435787, 871574, 1743148, 3486296
010100110010010110001
M39, 13466917 (12L-10S) 22 terms
3, 6, 12, 25, 51, 102, 205, 410, 821, 1643, 3287, 6575, 13151, 26302, 52605, 105210, 210420, 420841, 841682, 1683364, 3366729, 6733458
0011010111110100100101
M40, 20996011 (12L-11S) 23 terms
2, 5, 10, 20, 40, 80, 160, 320, 640, 1281, 2562, 5125, 10251, 20503, 41007, 82015, 164031, 328062, 656125, 1312250, 2624501, 5249002, 10498005
10000000101111110101011
M41, 24036583 (14L-9S) 23 terms
2, 5, 11, 22, 45, 91, 183, 366, 733, 1467, 2934, 5868, 11736, 23473, 46946, 93892, 187785, 375571, 751143, 1502286, 3004572, 6009145, 12018291
11011101100010011100111
M42, 25964951 (10L-13S) 23 terms
3, 6, 12, 24, 49, 99, 198, 396, 792, 1584, 3169, 6339, 12678, 25356, 50712, 101425, 202851, 405702, 811404, 1622809, 3245618, 6491237, 12982475
00011000011000110010111
M43, 30402457 (15L-8S) 23 terms
3, 7, 14, 28, 57, 115, 231, 463, 927, 1855, 3711, 7422, 14844, 29689, 59379, 118759, 237519, 475038, 950076, 1900153, 3800307, 7600614, 15201228
10011111110011110011001
M44, 32582657 (8L-15S) 23 terms
3, 7, 15, 31, 62, 124, 248, 497, 994, 1988, 3977, 7954, 15909, 31819, 63638, 127276, 254552, 509104, 1018208, 2036416, 4072832, 8145664, 16291328
11100010010110000000001
M45, 37156667 (16L-8S) 24 terms
2, 4, 8, 17, 35, 70, 141, 283, 566, 1133, 2267, 4535, 9071, 18142, 36285, 72571, 145143, 290286, 580572, 1161145, 2322291, 4644583, 9289166, 18578333
001101101111011100111011
M46, 42643801 (11L-13S) 24 terms
2, 5, 10, 20, 40, 81, 162, 325, 650, 1301, 2602, 5205, 10411 ,20822, 41644, 83288, 166577, 333154, 666309, 1332618, 2665237, 5330475, 10660950, 21321900
10001010101100010101011001
M47, 43112609 (10L-14S) 24 terms

2, 5, 10, 20, 41, 82, 164, 328, 657, 1315, 2631, 5262, 10525, 21051, 42102, 84204, 168408, 336817, 673634, 1347269, 2694538, 5389076, 10778152, 21556304
10010001110110001010000 1
M48, 57885161 (12L- 12S) 24 terms
3, 6, 13, 27, 55, 110, 220, 441, 883, 1766, 3533, 7066, 14132, 28264, 56528, 113056, 226113, 452227, 904455, 1808911, 3617822, 7235645, 14471290, 28942580
011100110100000111101001
M49, 74207281 (9L-16S) 25 terms
2, 4, 8, 17, 35, 70, 141, 283, 566, 1132, 2264, 4529, 9058, 18117, 36234, 72468, 144936, 289872, 579744, 1159488, 2318977, 4637955, 9275910, 18551820, 37103640
0011011000101000000110001
M50, 77232917 (13L-12S) 25 terms
2, 4, 9, 18, 36, 73, 147, 294, 589, 1178, 2356, 4713, 9427, 18855, 37711, 75422, 150845, 301691, 603382, 1206764, 2413528, 4827057, 9654114, 19308229, 38616458
0100110100111101100010101
M51, 82589933 (14L-11S) 25 terms
2, 4, 9, 19, 39, 78, 157, 315, 630, 1260, 2520, 5040, 10081, 20163, 40327, 80654, 161308, 322616, 645233, 1290467, 2580935, 5161870, 10323741, 20647483, 41294966
0111011000011100011101101

M52, 136279841 (13L-13S) 26 terms
2, 4, 8, 16, 32, 64, 129, 259, 519, 1039, 2079, 4158, 8317, 16635, 33271, 66542, 133085, 266171, 532343, 1064686, 2129372, 4258745, 8517490, 17034980, 34069960, 68139920
00000111110111011100100001

Terms pattern:

1,1,1,1, 2, 3,3,3, 4, 5,5,5, 8,8, 9, 10,10,10, 11,11. 12,12,12, 13,13,13, 14, 15,15, 16,16, 18,18, 19,19, 20,20, 21, 22, 23,23,23,23,23, 24,24,24,24, 25,25,25, 26.

BRANCHING:

2

M1, M3, M5, M6, M7, M10, M13, M14, M15, M16, M17, M19, M20, M21, M22, M23, M24, M25, M26, M27, M28, M30, M32, M34, M35, M36, M37, M40, M41, M45, M46, M47, M49, M50, M51, M52
(2**) M3**

(2, 4)

M6, M13, M15, M16, M17, M19, M20, M30, M45, M49, M52

(2, 4, 8**)**

M6, M16, M17, M20, M45, M49

(2, 4, 8, 17**)**

M16, M20

(2, 4, 8, 17, 34**) M16**

(2, 4, 8, 17, 34**) M20**

(2, 4, 8, 17**)**

M17, M45, M49

(2, 4, 8, 17, 35**) M17**,

(2, 4, 8, 17, 35, 70, 141, 283**)** M45, M49

(2, 4, 8, 17, 35, 70, 141, 283, 566**) M45**

(2, 4, 8, 17, 35, 70, 141, 283, 566**) M49**

(2, 4, 8)
M13, M19, M30, M52

(2, 4, 8, 15) M13
(2, 4, 8, 16) M19
(2, 4, 8, 16) M30, M52

(2, 4, 8, 16, 32) M4
(2, 4, 8, 16, 32) M30, M52

(2, 4, 8, 16, 32, 64) M30
(2, 4, 8, 16, 32, 64) M52

(2, 4,)
M7, M14, M15, M21, M22, M24, M34, M50, M51

(2, 4, 9)
M7, M15, M22, M24, M34, M51

(2, 4, 9, 19) M15, M51

(2, 4, 9, 19, 39) M15
(2, 4, 9, 19, 39) M51

(2, 4, 9, 19) M22, M24, M34

(2, 4, 9, 19, 38, 77, 155) M22
(2, 4, 9, 19, 38, 77, 155) M24
(2, 4, 9, 19, 38) M34

(2, 4, 9)
M14, M21, M50

(2, 4, 9, 18) M14, M21

(2, 4, 9, 18, 37, 75, 151) M14
(2, 4, 9, 18, 37, 75, 151) M21

(2, 4, 9, 18) M50

(2, 5)
M10, M23, M25, M26, M27, M28, M32, M35, M36, M37, M40, M41, M46, M47

(2, 5)
M10, M26, M32, M36, M37, M41

(2, 5, 11)
M10, M32, M37

(2, 5, 11, 22, 44) **M10**
(2, 5, 11, 23, 46, 92)
M32, M37
(2, 5, 11, 23, 46, 92, 184) **M32**
(2, 5, 11, 23, 46, 92, 184) **M37**
(2, 5, 11)
M26, M36, M41,
(2, 5, 11, 22) M26, M36, M41
(2, 5, 11, 22, 45) M26, M36
(2, 5, 11, 22, 45, 90, 181) **M26**
(2, 5, 11, 22, 45, 90, 181) **M36**
(2, 5, 11, 22, 45) **M41**

(2, 5)
M23, M25, M27, M28, M35, M40, M46, M47
(2, 5, 10)
(2, 5, 10, 21) M23, M27
(2, 5, 10, 21, 43) **M23**
(2, 5, 10, 21, 43) **M27**
(2, 5, 10, 21) M25, M28, M35
(2, 5, 10, 21, 42, 84) **M25**
(2, 5, 10, 21, 42, 84) **M28**
(2, 5, 10, 21, 42, 85) **M35**
(2, 5, 10, 20) **M40, M46**
(2, 5, 10, 20, 40) **M40**
(2, 5, 10, 20, 40) **M46**
(2, 5, 10, 20) **M47**

3
M2, M4, M5, M8, M9, M11, M12, M18, M29, M31, M33, M38, M39, M42, M43, M44, M48
(3) **M4**

(3, 6)
M5, M11, M29, M31, M33, M38, M48
(3, 6, 13) M11, M29, M31, M33, M38
(3, 6, 13, 26) M11, M29, M38

(3, 6, 13, 26, 53) **M11**
(3, 6, 13, 26, 53, 107) **M29**
(3, 6, 13, 26, 53) **M38**

(3, 6, 13, 26) M31, M33
(3, 6, 13, 26, 52) **M31**
(3, 6, 13, 26, 52) **M33**

(3, 6, 13) **M48**

(3, 6)

M18, M39, M42

(3, 6, 12) M18, M39
(3, 6, 12, 25) **M18**
(3, 6, 12, 25) **M39**

(3, 6, 12) **M42**

(3, 7)

M8, M9, M12, M44

(3, 7, 15) **M8**
(3, 7, 15, 31) **M44**
(3, 7, 15, 30) **M9**
(3, 7, 15, 31, 63) **M12**
(3, 7, 14) **M43**

Mersennes Exponents : Common terms from start

M1 2
M2 3
M3 (2)
M4 (3)
-
M5 (3, 6)

M5, M11, M29, M31, M33, M38, M48 - (2)

M6 (2, 4, 8)

M6, M16, M17, M20, M45, M49 - (3)

M7 (2, 4, 9)

M7, M15, M22, M24, M34, M51 - (3)

M8 (3, 7, 15)

M8, M9, M12, M44 – (2)

M8, M12, M44 - (3)

M9 (3, 7, 15, 30)

M8, M9, M12, M44 - (2)

M10 (2, 5, 11, 22, 44)

-

M11 (3, 6, 13, 26, 53)

M5

M11, M31, M33, M38, M48 – (3)

M11, M29, M38 – (4)

M11, M29 - (5)

M12 (3, 7, 15, 31, 63)

M8

M8, M9, M12, M44 – (2)

M8, M12, M44 - (3)

M13 (2, 4, 8, 15, 32, 65, 130, 260)

M13, M19, M30, M46, M52 - (3)

M14 (2, 4, 9, 18, 37, 75, 151, 303)

M14, M21, M50 – (3)

M14, M21 - (4)

-

M15 (2, 4, 9, 19, 39, 79, 159, 319, 639)

M7

M15, M51 - (4)

M16 (2, 4, 8, 17, 34, 68, 137, 275, 551, 1101)

M6

M20 - (4)

M17 (2, 4, 8, 17, 35, 71, 142, 285, 570, 1140)

M6

M17, M45, M49 - (4)

M18 (3, 6, 12, 25, 50, 100, 201, 402, 804, 1608)

M18, M39 - (3)

M19 (2, 4, 8, 16, 33, 66, 132, 265, 531, 1063, 2126)

M13, M19, M30, M46, M52 – (3)

M20 (2, 4, 8, 17, 34, 69, 138, 276, 552, 1105, 2211)

M6

M20 -(4)

M21 (2, 4, 9, 18, 37, 75, 151, 302, 605, 1211, 2422, 4844)

M14, M21, M50 – (3)

M14, M21 - (4)

M22 (2, 4, 9, 19, 38, 77, 155, 310, 621, 1242, 2485, 4970)

M7

M22, M24, M34 - (4)

-

M23 (2, 5, 10, 21, 43, 87, 175, 350, 700, 1401, 2803, 5606)

M23, M27 - (4)

M24 (2, 4, 9, 19, 38, 77, 155, 311, 623, 1246, 2492, 4984, 9968)

M7

M25 (2, 5, 10, 21, 42, 84, 169, 339, 678, 1356, 2712, 5425, 10850)

M25, M28, M35 - (4)

M26 (2, 5, 11, 22, 45, 90, 181, 362, 725, 1450, 2901, 5802, 11604)

M26, M36, M41 – (3)

M26, M36 - (6)

M27 (2, 5, 10, 21, 43, 86, 173, 347, 695, 1390, 2781, 5562, 11124, 22248)

M23, M27 - (4)

M28 (2, 5, 10, 21, 42, 84, 168, 336, 673, 1347, 2695, 5390, 10780, 21560, 43121)

M25, M28, M35 - (4)

-

M29 (3, 6, 13, 26, 53, 107, 215, 431, 863, 1726, 3453, 6906, 13812, 27625, 55251)

M11

M5

M11, M29, M38 – (4)

M11, M29, - (5)

M30 (2, 4, 8, 16, 32, 64, 128, 257, 515, 1031, 2063, 4126, 8253, 16506, 33012, 66024)

M13, M19, M30, M46, M52 – (3)

M46, M30, M52 – (4)

M30, M52 - (5)

M31 (3, 6, 13, 26, 52, 105, 211, 422, 844, 1688, 3376, 6752, 13505, 27011, 54022, 108045)

M5

M11, M31, M33, M38, M48 – (3)

M31, M33 - (4)

M32 (2, 5, 11, 23, 46, 92, 184, 369, 739, 1478, 2956, 5912, 11825, 23651, 47302, 94604, 189209, 378419)

M32, M37 - (6)

M33 (3, 6, 13, 26, 52, 104, 209, 419, 839, 1678, 3357, 6714, 13428, 26857, 53714, 107429, 214858, 429716)

M5

M11, M31, M33, M38, M48 – (3)

M31, M33 - (4)

-

M34 (2, 4, 9, 19, 38, 76, 153, 307, 614, 1228, 2456, 4913, 9826, 19652, 39305, 78611, 157223, 314446, 628893)

M7

M35 (2, 5, 10, 21, 42, 85, 170, 341, 682, 1365, 2730, 5461, 10923, 21847, 43695, 87391, 174783, 349567, 699134)

M25, M28, M35 - (4)

M36 (2, 5, 11, 22, 45, 90, 181, 363, 726, 1453, 2906, 5812, 11625, 23251, 46503, 93006, 186013, 372027, 744055, 1488110)

M26, M36, M41 – (3)

M26, M36 - (6)

M37 (2, 5, 11, 23, 46, 92, 184, 368, 737, 1475, 2950, 5901, 11802, 23604, 47209, 94418, 188836, 377672, 755344, 1510688)

M32, M37 - (6)

M38 (3, 6, 13, 26, 53, 106, 212, 425, 851, 1702, 3404, 6809, 13618, 27236, 54473, 108946, 217893, 435787, 871574, 1743148, 3486296)

M11

M5

M11, M31, M33, M38, M48 – (3)

M11, M29, M38 - (4)

-

M39 (3, 6, 12, 25, 51, 102, 205, 410, 821, 1643, 3287, 6575, 13151, 26302, 52605, 105210, 210420, 420841, 841682, 1683364, 3366729, 6733458)

M18, M39 - (3)

M40 (2, 5, 10, 20, 40, 80, 160, 320, 640, 1281, 2562, 5125, 10251, 20503, 41007, 82015, 164031, 328062, 656125, 1312250, 2624501, 5249002, 10498005)

M40, M47 - (3)

M41 (2, 5, 11, 22, 45, 91, 183, 366, 733, 1467, 2934, 5868, 11736, 23473, 46946, 93892, 187785, 375571, 751143, 1502286, 3004572, 6009145, 12018291)

M26, M36, M41 - (3)

M42 (3, 6, 12, 24, 49, 99, 198, 396, 792, 1584, 3169, 6339, 12678, 25356, 50712, 101425, 202851, 405702, 811404, 1622809, 3245618, 6491237, 12982475)

UNIQUE

M43 (3, 7, 14, 28, 57, 115, 231, 463, 927, 1855, 3711, 7422, 14844, 29689, 59379, 118759, 237519, 475038, 950076, 1900153, 3800307, 7600614, 15201228)

UNIQUE

M44 (3, 7, 15, 31, 62, 124, 248, 497, 994, 1988, 3977, 7954, 15909, 31819, 63638, 127276, 254552, 509104, 1018208, 2036416, 4072832, 8145664, 16291328)

M12

M8, M9, M12, M44 – (2)

M8, M12, M44 - (3)

M45 (2, 4, 8, 17, 35, 70, 141, 283, 566, 1133, 2267, 4535, 9071, 18142, 36285, 72571, 145143, 290286, 580572, 1161145, 2322291, 4644583, 9289166, 18578333)

M6

M17, M45, M49 – (4)

M45, M49 – (8)

-**M46** (2, 5, 10, 20, 40, 81, 162, 325, 650, 1301, 2602, 5205, 10411 ,20822, 41644, 83288, 166577, 333154, 666309, 1332618, 2665237, 5330475, 10660950, 21321900)

M13, M19, M30, M46, M52 – (3)

M46, M30, M52 - (4)

M47 (2, 5, 10, 20, 41, 82, 164, 328, 657, 1315, 2631, 5262, 10525, 21051, 42102, 84204, 168408, 336817, 673634, 1347269, 2694538, 5389076, 10778152, 21556304)

M40, M47 - (3)

M48 (3, 6, 13, 27, 55, 110, 220, 441, 883, 1766, 3533, 7066, 14132, 28264, 56528, 113056, 226113, 452227, 904455, 1808911, 3617822, 7235645, 14471290, 28942580)

M5

M11, M31, M33, M38, M48 - (3)

M49 (2, 4, 8, 17, 35, 70, 141, 283, 566, 1132, 2264, 4529, 9058, 18117, 36234, 72468, 144936, 289872, 579744, 1159488, 2318977, 4637955, 9275910, 18551820, 37103640)

M6

M17, M45, M49 – (4)

M45, M49 – (8)

M50 (2, 4, 9, 18, 36, 73, 147, 294, 589, 1178, 2356, 4713, 9427, 18855, 37711, 75422, 150845, 301691, 603382, 1206764, 2413528, 4827057, 9654114, 19308229, 38616458)

M14, M21, M50 - (3)

M51 (2, 4, 9, 19, 39, 78, 157, 315, 630, 1260, 2520, 5040, 10081, 20163, 40327, 80654, 161308, 322616, 645233, 1290467, 2580935, 5161870, 10323741, 20647483, 41294966**)**

M7

M15, M51 - (4)

-

M52 (2, 4, 8, 16, 32, 64, 129, 259, 519, 1039, 2079, 4158, 8317, 16635, 33271, 66542, 133085, 266171, 532343, 1064686, 2129372, 4258745, 8517490, 17034980, 34069960, 68139920**)**

M13, M19, M30, M46, M52 – (3)

M46, M30, M52 – (4)

M30, M52 - (5)

Conclusions

1.The Mersennes exponents are results of latices recursive sequences where next terms is in form of previous term x2 or x3+1.

2. The Mersennes exponents numbers always end in previous numbers in form of previous term x2+1.

3. The Mersennes exponents sequences latices start with M1= 2 or M2= 3 as two separate BASES (the first two known Mersennes Exponents).

4. The Mersennes exponents sequences latices terms as x2 or x2+1 is determinant by unique terms. Duplicate terms are growing from the base.

More study are necessarily.

Available app in github.

Appendix:

Base 2	**Base 3**
M1	M2
M3	
	M4
	M5
M6	
M7	
	M8
	M9
M10	
	M11
	M12
M13	
M14	
M15	
M16	
M17	
	M18
M19	
M20	
M21	
M22	
M23	
M24	
M25	
M26	

M27
M28
M29
M30
M31
M32
M33
M34
M35
M36
M37
M38
M39
M40
M41
M42
M43
M44
M45
M46
M47
M48
M49
M50
M51
M52

Reference:

(1) OEIS
(2) Wikipedia